Normal Operators

Def.: $(V, \langle , \rangle)$ is $\mathbb{C}$- $\sim$

$T: V \to V$ is normal

$\iff T^*T = TT^*$

E.g.: T is self-adjoint

$\Rightarrow T^* = T$

$\Rightarrow T^*T = T^2 = TT^*$

$\Rightarrow T$ is normal

E.g. 2: T is unitary

$\Rightarrow T^*T = I$

$$\begin{cases} \underbrace{\dim(V) < \infty} \\ T \text{ is unitary} \end{cases} \Rightarrow T^{-1} = T^*$$

$\Rightarrow TT^* = I$

$\Rightarrow T^*T = I = TT^*$

Remark:

$\dim(V) = \infty$

T is unitary $\iff T^*T = TT^* = I$

$\dim(V) < \infty$

T is unitary $\iff T^*T = I \Rightarrow TT^* = I$

Hence T is unitary $\Rightarrow \forall \dim(V)$, T is normal

Theorem:

$\left.\begin{array}{l}(V, \langle\,,\,\rangle) \text{ is finite dimensional} \\ T: V \to V \text{ is normal}\end{array}\right\} \Rightarrow \exists$ orthogonal basis of eigenvectors of T

Prop:

T is normal

$\Rightarrow$ { (1) ⋮ (2) } (见书)

Proof (of Prop.):

(1) $\|T(\underline{v})\|^2 = \langle T(\underline{v}), T(\underline{v})\rangle$

$= \langle T^* T(\underline{v}), \underline{v}\rangle$

[T is normal] $= \langle T T^*(\underline{v}), \underline{v}\rangle$

$= \langle \underline{v}, T T^*(\underline{v})\rangle$

$= \langle T^*(\underline{v}), T^*(\underline{v})\rangle$

$= \|T^*(\underline{v})\|$

(2) $(T-\lambda I)^*(T-\lambda I)$

$=(T^*-\overline{\lambda}I)(T-\lambda I)$

$=T^*T-\lambda T^*-\overline{\lambda}T+\overline{\lambda}\lambda I$

(T is normal) $=TT^*-\lambda T^*-\overline{\lambda}T+\overline{\lambda}\lambda I$

$=(T-\lambda I)(T^*-\overline{\lambda}I)$

$=(T-\lambda I)(T-\lambda I)^*$

(3) $T(\underline{v})=\lambda\underline{v}$

$\Leftrightarrow (T-\lambda I)\underline{v}=\underline{0}$

$\Leftrightarrow \|(T-\lambda I)\underline{v}\|^2=0$

($(T-\lambda I)$ is normal) $\Leftrightarrow \|(T-\lambda I)^*\underline{v}\|^2=0$

$\Leftrightarrow \|(T^*-\overline{\lambda}I)\underline{v}\|^2=0$

$\Leftrightarrow (T^*-\overline{\lambda}I)\underline{v}=0$

$\Leftrightarrow$ ~~~

(4) (只需)

(4) $\langle T^*(\underline{v}),\underline{w}\rangle=\langle\underline{v},T(\underline{w})\rangle=\langle\underline{v},\mu\underline{w}\rangle$

$=\mu\langle\underline{v},\underline{w}\rangle$

// (3)

$\langle\overline{\lambda}\underline{v},\underline{w}\rangle=(\overline{\lambda})\langle\underline{v},\underline{w}\rangle=\lambda\langle\underline{v},\underline{w}\rangle$

$\therefore \lambda\langle\underline{v},\underline{w}\rangle=\mu\langle\underline{v},\underline{w}\rangle$

$\Rightarrow (\lambda-\mu)\langle\underline{v},\underline{w}\rangle=0$

$\neq 0$

$\Rightarrow \langle\underline{v},\underline{w}\rangle=0$

Proof (of Theorem):

Use induction on dim. of V

When $\dim(V) = 1$:

Take any ($\forall$) unit vector in V in a basis of V

Assume that:

For $\forall$ v. space V_k where $\dim(V_k) = k$

and $\forall$ normal operator $T_k : V_k \to V_k$,

the thm. is true

Assume V_{k+1} is a v. space and $\dim(V_{k+1}) = k+1$

and $T_{k+1} : V_{k+1} \to V_{k+1}$ is a normal operator

(WTS: $\exists$ orthogonal basis

of e.vectors of T_{k+1} in V_{k+1})

Claim 1: T_{k+1} has an e.vector $\underline{u}$ ($\exists$)

($\Rightarrow \forall \underline{w} \in U$ is e.vector of T_{k+1})

Then take $U := \mathrm{Span}\{\underline{u}\}$ ($\dim(U) = 1$)

and consider $U^\perp$ ($\dim(U^\perp) = k$) s.t. $V = U \oplus U^\perp$

Claim 2: $U^\perp \leq V$ is T-invariant

Proof: Take $\forall \underline{w} \in U^\perp$

Then $\forall \underline{u}' \in U$, $\langle \underline{u}', T(\underline{w}) \rangle$

$= \langle T^*(\underline{u}'), \underline{w} \rangle$

$\in U$, $\in U^\perp$

(Prop. (3)) $\to = \langle \bar{\lambda}\, \underline{u}', \underline{w} \rangle = \lambda \langle \underline{u}', \underline{w} \rangle = 0$

$\Rightarrow T(\underline{w}) \in U^\perp$

Claim 3:

Corollary (Spectral Theorem):

$(V, \langle , \rangle)$ is $\mathbb{C}$-~

$T: V \to V$ is normal

$\Rightarrow \exists$ operators $P_1, \cdots, P_k$,

s.t. { $\forall i$, P_i is self-adjoint $(P_i^* = P_i)$ ~(a)

$\forall i$, P_i is idempotent $(P_i^2 = P_i)$ ~(b)

$\forall i \neq j$, $P_i P_j = 0$ ~(c)

$\sum_{i=1}^{k} P_i = (P_1 + \cdots + P_k =) I$ ~(d)

and $T = \lambda_1 P_1 + \cdots + \lambda_k P_k$,

where λ_i's are e.values of T

Proof:

T is normal

(前一个Thm.) $\Rightarrow$ T has a basis ① of e.vectors ②

$\Rightarrow$ T is diagonable ★

$\Rightarrow m_T(x) = (x-\lambda_1) \cdots (x-\lambda_k)$,

where λ_i's are the e.values of T and are distinct

② Recall the proof of PP T

$(\forall i, V_i = \ker(T - \lambda_i I) \Rightarrow V = V_1 \oplus \cdots \oplus V_k)$

let $q_i(x) = \frac{m_T(x)}{(x-\lambda_i)}$ $(= (x-\lambda_1) \cdots (x-\lambda_{i-1})(x-\lambda_{i+1}) \cdots (x-\lambda_k))$

$\Rightarrow \gcd(q_1(x) \cdots q_k(x)) = 1$

(Beazout's) $\Rightarrow \exists a_1(x), \cdots, a_k(x)$, s.t. $a_1(x)q_1(x) \cdots a_k(x)q_k(x) = 1$

Take $P_i := a_i(T)\,q_i(T)$,

then $P_1 + \cdots + P_k = I$ $\xrightarrow{\text{(d)}}$

$\left(\Rightarrow \underbrace{P_1(\underline{v})}_{\in V_1} + \cdots + \underbrace{P_k(\underline{v})}_{\in V_k} = \underbrace{\underline{v}}_{\in V}\right)$

If $i \neq j$, then

$P_i P_j = a_i(T)\,q_i(T)\,a_j(T)\,q_j(T)$

$= R_i(T)\,a_j(T)\,q_i(T)\,q_j(T)$

$\overset{(***)}{=} q_i(T)\,a_j(T) \cdots m_T(T) = \underline{0}$

$P_i = P_i I$

$\xrightarrow{\text{(d)}} = P_i(P_1 + \cdots + P_k)$

$\xrightarrow{\text{(c)}} = P_i P_i = P_i^2 \xrightarrow{\text{(b)}}$)

$\forall \underline{v} \in V$,

$T(\underline{v}) = T \circ I(\underline{v})$

$= T(P_1 + \cdots + P_k)(\underline{v})$

$= T(P_1(\underline{v})) + \cdots + T(P_k(\underline{v}))$

$\in V_1 = \ker(T - \lambda_1 I)$ $\quad \in V_k = \ker(T - \lambda_k I)$

($(T - \lambda_i I)(P_i(\underline{v})) = T(P_i(\underline{v})) - \lambda_i P_i(\underline{v}) = 0$

$\Rightarrow T(P_i(\underline{v})) = \lambda_i P_i(\underline{v})$)

$= \lambda_1 P_1(\underline{v}) + \cdots + \lambda_k P_k(\underline{v})$

(接下頁)

$T(v) = (\lambda_1 P_1 + \cdots + \lambda_k P_k)(v) \quad \forall v \in V$ (5) ✓

So we're left with (1).

Lemma $T: V \to V$ is normal $\Leftrightarrow$ $T^* = p(T)$ for some poly. $p(x) \in \mathbb{F}[x]$

Assuming Lemma, then $\left(P_i = a_i(T) q_i(T) = b_n T^n + \cdots + b_1 T + b_0 I\right.$

$P_i^* = \overline{b_n}(T^*)^n + \cdots + \overline{b_1}(T^*) + \overline{b_0} I$

(续⑦)